

Cloud Computing – advanced topics

In this assignment, each group must select a topic from the list below and prepare a report covering the key elements of their study. Whenever possible, the study should include a prototype or practical experiment related to the chosen topic (consult before deployment), as well as an analysis of development costs and operating costs.

List of topics for in-depth analysis:

- *Advanced DevOps Pipelines*
- *Serverless Edge Computing & IoT Data Pipelines*
- *Cloud Orchestration and Automation*
- *Combining on-premises with cloud-based storage*
- *Cloud Cost Optimization & FinOps Automation*
- *Cloud Security (Data, Process, System)*
- *Big Data processing frameworks: Hadoop and Spark*
- *Serverless Architecture Design*
- *AI-driven cloud resource allocation*
- *Virtualization and creation of resources to offer private cloud services*
- *Autoscaling management when instances need to maintain a state*

Submission

Project assignments must be submitted by uploading a .zip file in the designated area on Moodle. Students may work individually or in pairs.

Submission deadline

refer to Moodle